

# YICHENG XU

Mechanical Engineer

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📍 Irvine, CA

📁 Portfolio

## SUMMARY

I am a Ph.D. student at UC Irvine studying Systems and Control under the supervision of Professor Faryar Jabbari. I have an education background in mechanical engineering, especially on optimization and control theory. My current interest is multi-agent systems, anti-windup control, event-triggered control, linear parameter varying systems, distributed optimization on graphs, and control barrier functions.

## EDUCATION

|                     |                                                                                                             |           |
|---------------------|-------------------------------------------------------------------------------------------------------------|-----------|
| Sep 2021 - now      | <b>Ph.D. candidate</b> , Mechanical Engineering<br>University of California, Irvine                         | 3.941/4.0 |
| Sep 2020 - Jun 2021 | <b>Master of Science</b> , Mechanical Engineering<br>University of California, Irvine                       | 4.0/4.0   |
| Sep 2016 - Jun 2020 | <b>Bachelor of Science</b> , Automation<br>Southeast University<br>Exchange at UC Irvine in the fourth year | 3.6/4.0   |

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                         |                                 |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------|
| <b>Mechanical Engineer   UCI's Engineering Design in Industry Program</b>                                                                                                                                                                                                                                                                                               | Arduino, Solidworks, 3D print   | Jan 2020 - Mar 2020 |
| <ul style="list-style-type: none"><li>Lead development of innovative personal hygiene device for hot, damp, and sanitized towelette delivery</li><li>Collaborate with cross-functional team to optimize product functionality and user experience</li><li>More information on <a href="https://ethanyxu.com/WhoopyWipes">https://ethanyxu.com/WhoopyWipes</a></li></ul> |                                 |                     |
| <b>Mechanical Engineer   UCI's Engineering Design in Industry Program</b>                                                                                                                                                                                                                                                                                               | Signal processing, Solidworks   | Sep 2019 - Dec 2019 |
| <ul style="list-style-type: none"><li>Develop hands-free, noninvasive clinical solution for enhanced blood vessel visualization</li><li>Create comprehensive Bill of Materials (BOM) for efficient product manufacturing</li><li>Apply principles of objective and quantifiable assessment in medical device design</li></ul>                                           |                                 |                     |
| <b>Strategy consulting Intership   PricewaterhouseCoopers</b>                                                                                                                                                                                                                                                                                                           | Strategy consult, Communication | Sep 2017 - Dec 2017 |
| <ul style="list-style-type: none"><li>Provide strategic consulting services to a prominent Electric and Electronic Manufacturing company</li><li>Utilize enterprise analysis techniques to identify market opportunities</li></ul>                                                                                                                                      |                                 |                     |

## ACADEMIC EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------|
| <b>Graduate Student Lead   Hacker Fab at UC Irvine</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mentoring, Photolithography, PCB | Sep 2024 - Now |
| <ul style="list-style-type: none"><li>Lead interdisciplinary research team at UCI's branch of Hacker Fab (<a href="https://ethanyxu.com/IrvineHackerFab">https://ethanyxu.com/IrvineHackerFab</a>)</li><li>Spearhead development of optical lithography projects, e.g., tube furnace, spin coater, and patterning.</li><li>Drive fund-raising initiatives and oversee project management. Cultivate collaborative environment as graduate student leader (<a href="https://ethanyxu.com/UROP">https://ethanyxu.com/UROP</a>)</li></ul> |                                  |                |
| <b>Teaching Assistant / Instructor   University of California, Irvine</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Teaching, Communication          | Sep 2021 - Now |
| <ul style="list-style-type: none"><li>Assist Professor Faryar Jabbari in teaching ENGRMAE 80: Dynamics at University of California, Irvine</li><li>Deliver one-on-one support to enhance student understanding of complex dynamics concepts</li><li>Evaluate and grade homework assignments and exams, ensuring fair assessment</li></ul>                                                                                                                                                                                              |                                  |                |

## AWARDS

|                                                                                                                                                                                                                                      |                                  |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------|
| <b>CALIT2 IRT Graduate Student Mentor Stipend, University of California, Irvine</b>                                                                                                                                                  | Mentoring, Project Management    | Sep 2025 - May 2026 |
| <ul style="list-style-type: none"><li>Awarded a \$1,000 UCI stipend as graduate student mentor in the CALIT2 IRT Program</li><li>Coordinating project direction and cross-disciplinary collaboration</li></ul>                       |                                  |                     |
| <b>Provincial Third Prize for National College Mathematical Contest in Modeling</b>                                                                                                                                                  | Modelling, Optimisation          | Sep 2018 - Dec 2018 |
| <ul style="list-style-type: none"><li>Lead team in designing regulation protocol for RGV robots in automated warehouse systems</li><li>Optimize working patterns to minimize robot stop time, enhancing overall efficiency</li></ul> |                                  |                     |
| <b>Excellence Award for 20th electronic design contest of Southeast University</b>                                                                                                                                                   | MCU development(STM32), Circuits | Apr 2018 - Jun 2018 |
| <ul style="list-style-type: none"><li>Collaborate in team to design feedback control system for DC power under varying load conditions</li><li>Test on bread board with oscilloscope, finalize with PCB board design</li></ul>       |                                  |                     |
| <b>First price for 14th RoboCup of Southeast University</b>                                                                                                                                                                          | ROS, Python, Pattern recognition | Dec 2017 - Jan 2018 |
| <ul style="list-style-type: none"><li>Program a robot using ROS for navigation and data collection</li><li>Use OpenCV in Python to recognize and track objects</li></ul>                                                             |                                  |                     |

PUBLICATIONS AND PATENTS

[1] P. Ong, Y. Xu, R. M. Bena, F. Jabbari, and A. D. Ames, "Matrix Control Barrier Functions," *IEEE Transactions on Automatic Control*, pp. 1–13, Aug. 2026. doi: 10.1109/TAC.2026.3701302 Accessed: Jun. 10, 2026.

[2] Y. Xu and F. Jabbari, "Distributed Optimization of Finite Condition Number for Laplacian Matrix in Multi-Agent Systems," *International Journal of Robust and Nonlinear Control*, vol. 36, no. 9, pp. 5030–5043, Jun. 2026. doi: 10.1002/rnc.70479

[3] Y. Xu and F. Jabbari, "Discrete-Time Leader-Following Multiagent Systems: Saturation Constraints and Event-Triggered Control," *IEEE Transactions on Control of Network Systems*, vol. 12, no. 2, pp. 1354–1368, Jun. 2025, doi: 10.1109/TCNS.2024.3516578

[4] Y. Xu and F. Jabbari, "Distributed Optimization of Network Weights for Improved Performance," in *2024 American Control Conference (ACC)*, IEEE, Jul. 2024, pp. 1392–1397. doi: 10.23919/ACC60939.2024.10644563

[5] Y. Xu and F. Jabbari, "Spline-based parameter varying output feedback synthesis with improved  $L_2$  gain," in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, IEEE, Dec. 2024, pp. 1455–1460. doi: 10.1109/CDC56724.2024.10886461

[6] Y. Xu and F. Jabbari, "Discrete-Time Output Feedback under Bounded Actuators: Single and Multi-agent Problems," in *2022 IEEE 61st Conference on Decision and Control (CDC)*, IEEE, Dec. 2022, pp. 4865–4871. doi: 10.1109/CDC51059.2022.9992896

[7] Y. Xu, Y. Chen, T. Liu, and W. Chen, "Optimization Algorithm for Power Flow Calculation Using Graph Theory," in *Lecture Notes in Electrical Engineering Proceedings of 2019 Chinese Intelligent Systems Conference*, 2020, pp. 142–150. doi: 10.1007/978-981-32-9698-5\_17

[8] Y. Chen, W. Chen, Y. Xu, and T. Liu, "A New Method of Optimizing Newton-Raphson Power Flow Based on Graph decomposition," Oct. 2019. Accessed: Jul. 11, 2019.

PRESENTATION

- 1. May 2026, "Control of Heterogeneous Multi-Agent Systems with Active Leader", Presented at 2026 American Control Conference (ACC), New Orleans, LA, USA Program
- 2. April 2025, "Distributed optimization of the finite condition number of the Laplacian matrix", Presented at 45th Southern California Control Workshop, San Diego, CA, USA Program
- 3. Dec 2024, "Spline-based parameter varying output feedback synthesis with improved  $L_2$  gain", Presented at 2024 IEEE 63rd Conference on Decision and Control (CDC), Milano, Italy Presented Paper
- 4. Jul 2024, "Distributed Optimization of Network Weights for Improved Performance", Presented at 2024 American Control Conference (ACC), Toronto, ON, Canada Presented Paper
- 5. Dec 2022, "Discrete-Time Output Feedback under Bounded Actuators: Single and Multi-agent Problems", Presented at 2022 IEEE 61st Conference on Decision and Control (CDC), Cancun, Mexico Presented Paper
- 6. Oct 2019, "Optimization Algorithm for Power Flow Calculation Using Graph Theory", Presented at 2019 Chinese Intelligent Systems Conference (CISC), Haikou, China Program

SKILLS

|                         |                                                                                                                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program Language:       | Matlab, Python, C/C++, Shell, LaTeX, HTML                                                                                                                                                           |
| Control Theory:         | PID Control, Nonlinear Control, Robust Control ( $H_\infty$ , $l_2$ gain and gain scheduling), Event-Triggered Control, Control Barrier Function, Multi-Agent System, Anti-Windup Control, MPC, LQR |
| Mechanical Engineering: | 3D printing, CAD(Solidworks), Lagrangian Mechanic, System Model via first principle, System Identification via Bode Plot, Linear Time-Invariant System,                                             |
| Electric Engineering:   | Micro-Controller Develop (Arduino, STM32, RaspberryPi), PCB desgin(Altium Designer, Easy EDA, KiCad), Oscilloscopes, Soldering, Circuits and Signals, Power and Energy System (AC,DC), DSP          |
| Optimization:           | Convex Optimization, Semi-definite Programming, Numerical Methods (Augmented Lagrangian, Interior Point Method, variations of Newton's method), ADMM                                                |

LANGUAGES

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|-----------|--------------------------|
| English:  | Professional proficiency |
| Mandarin: | Native speaker           |